I U S

Faculté des Sciences et des Technologies

Réseaux II

TD n° 7

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Objectifs

1. Configurer un réseau avec 3 routeurs

2. Mettre en place un Cisco ASA Firewall

3. Configurer le routage inter-sites (OSPF)

4. Créer une DMZ sécurisée

5. Implémenter des ACL sur ASA

6. Tester la sécurité du réseau

7. Configurer un VPN IPsec site-to-site

8. Sécuriser les communications entre 3 réseaux

9. Utiliser des routeurs Cisco (ou ASA si extension)

Comprendre :

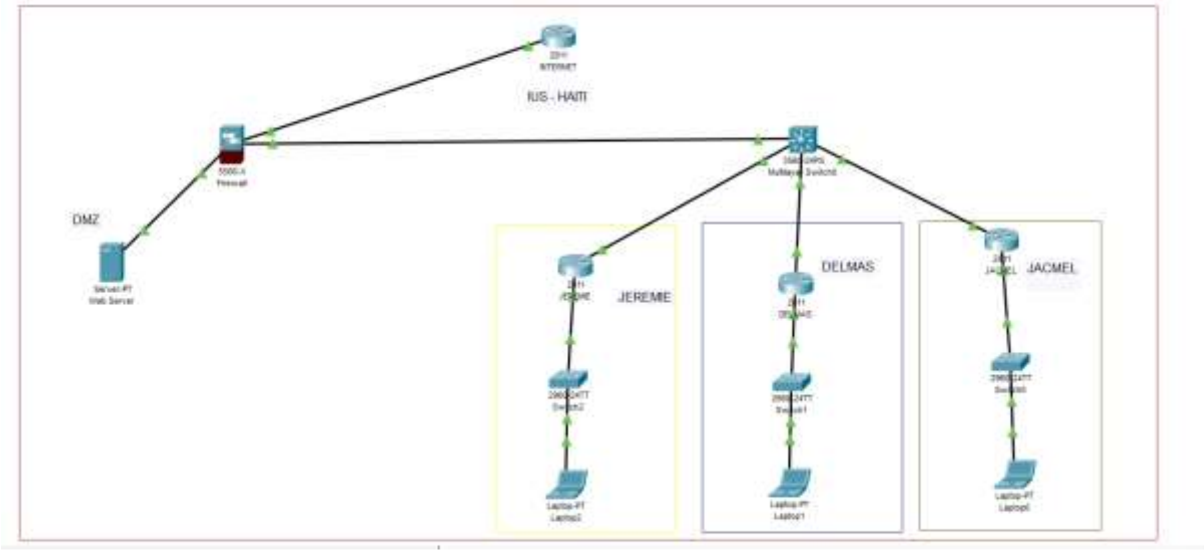
Phase 1 (ISAKMP)

Phase 2 (IPsec)

10. Tester le tunnel VPN

11. Vérifier le chiffrement

- **Création de la topologie**



Partie 1 - Sécurisation du réseau universitaire IUS avec Cisco ASA Firewall

- **Configuration Routeur Internet**

```
R0(config)#interface Fa0/0
R0(config-if)#ip address 203.0.113.1 255.255.255.252
R0(config-if)#no shutdown
R0(config-if)#interface Fa0/1
R0(config-if)#ip address 198.51.100.1 255.255.255.252
R0(config-if)# no shutdown

R0(config-if)#
%LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to down
ip routing
R0(config)#ip route 0.0.0.0 0.0.0.0 198.51.100.2
R0(config)#ip route 192.168.0.0 255.255.0.0 203.0.113.2
R0(config)#ip route 10.0.0.0 255.0.0.0 203.0.113.2
R0(config)#end
R0#
%SYS-5-CONFIG_I: Configured from console by console
```

- **Configuration Firewall**

ASA1

Physical Config CLI Attributes

IOS Command Line Interface

```
ciscoasa>en
Password:
ciscoasa#conf t
ciscoasa(config)#interface GigabitEthernet1/1
ciscoasa(config-if)#nameif outside
INFO: Security level for "outside" set to 0 by default.
ciscoasa(config-if)#security-level 0
ciscoasa(config-if)#ip address 203.0.113.2 255.255.255.252
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
ciscoasa(config-if)#no shutdown

%LINK-5-CHANGED: Interface GigabitEthernet1/1, changed state to down
ciscoasa(config-if)#interface GigabitEthernet1/2
ciscoasa(config-if)# nameif dmz
INFO: Security level for "dmz" set to 0 by default.
ciscoasa(config-if)#security-level 50
ciscoasa(config-if)#ip address 192.168.100.1 255.255.255.0
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
ciscoasa(config-if)#no shutdown

%LINK-5-CHANGED: Interface GigabitEthernet1/2, changed state to down
ciscoasa(config-if)#interface GigabitEthernet1/3
ciscoasa(config-if)#nameif inside
INFO: Security level for "inside" set to 100 by default.
ciscoasa(config-if)#security-level 100
ciscoasa(config-if)#ip address 10.0.0.1 255.255.255.0
Waiting for the earlier webvpn instance to terminate...
Previous instance shut down.Starting a new one
ciscoasa(config-if)#no shutdown

%LINK-5-CHANGED: Interface GigabitEthernet1/3, changed state to down
ciscoasa(config-if)#route outside 0.0.0.0 0.0.0.0 203.0.113.1
ciscoasa(config)#route inside 10.1.0.0 255.255.0.0 10.0.0.2
ciscoasa(config)#route inside 192.168.0.0 255.255.0.0 10.0.0.2
ciscoasa(config)#object network INSIDE-NET
```

```
ciscoasa(config-network-object)#subnet 192.168.0.0 255.255.0.0
ciscoasa(config-network-object)#nat (inside,outside) dynamic interface
ciscoasa(config-network-object)#object network WEB-SERVER
ciscoasa(config-network-object)#host 192.168.100.10
ciscoasa(config-network-object)#nat (dmz,outside) static 203.0.113.10
ciscoasa(config-network-object)#access-list INSIDE-IN extended deny ip 192.168.30.0
255.255.255.0 192.168.100.0 255.255.255.0
ciscoasa(config)#access-list INSIDE-IN extended permit ip any any
ciscoasa(config)#access-group INSIDE-IN in interface inside
ciscoasa(config)#access-list OUTSIDE-IN extended permit tcp any host 203.0.113.10 eq 80
ciscoasa(config)#access-list OUTSIDE-IN extended permit tcp any host 203.0.113.10 eq 443
ciscoasa(config)#access-group OUTSIDE-IN in interface outside
ciscoasa(config)#end
ciscoasa#
```

- **Configuration Multilayer Switch (OSPF)**

Multilayer Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#interface FastEthernet0/0
%Invalid interface type and number
Switch(config)#interface FastEthernet0/1
Switch(config-if)#no switchport
Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 10.0.0.2 255.255.255.252
Switch(config-if)#no shutdown
Switch(config-if)#interface FastEthernet0/2
Switch(config-if)#no switchport
Switch(config-if)#ip address 10.0.1.1 255.255.255.252
Switch(config-if)#no shutdown
Switch(config-if)#interface FastEthernet0/3
Switch(config-if)#no switchport
Switch(config-if)#ip address 10.0.2.1 255.255.255.252
Switch(config-if)#no shutdown
Switch(config-if)#interface FastEthernet0/4
Switch(config-if)#no switchport
Switch(config-if)#ip address 10.0.3.1 255.255.255.252
Switch(config-if)#no shutdown
Switch(config-if)#network 10.0.0.0 0.0.0.3 area 0
                    ^
% Invalid input detected at '^' marker.

Switch(config-if)#router ospf 1
Switch(config-router)#network 10.0.0.0 0.0.0.3 area 0
Switch(config-router)#network 10.0.1.0 0.0.0.3 area 0
Switch(config-router)#network 10.0.2.0 0.0.0.3 area 0
Switch(config-router)#network 10.0.3.0 0.0.0.3 area 0
Switch(config-router)#ip route 0.0.0.0 0.0.0.0 10.0.0.1
Switch(config)#
```

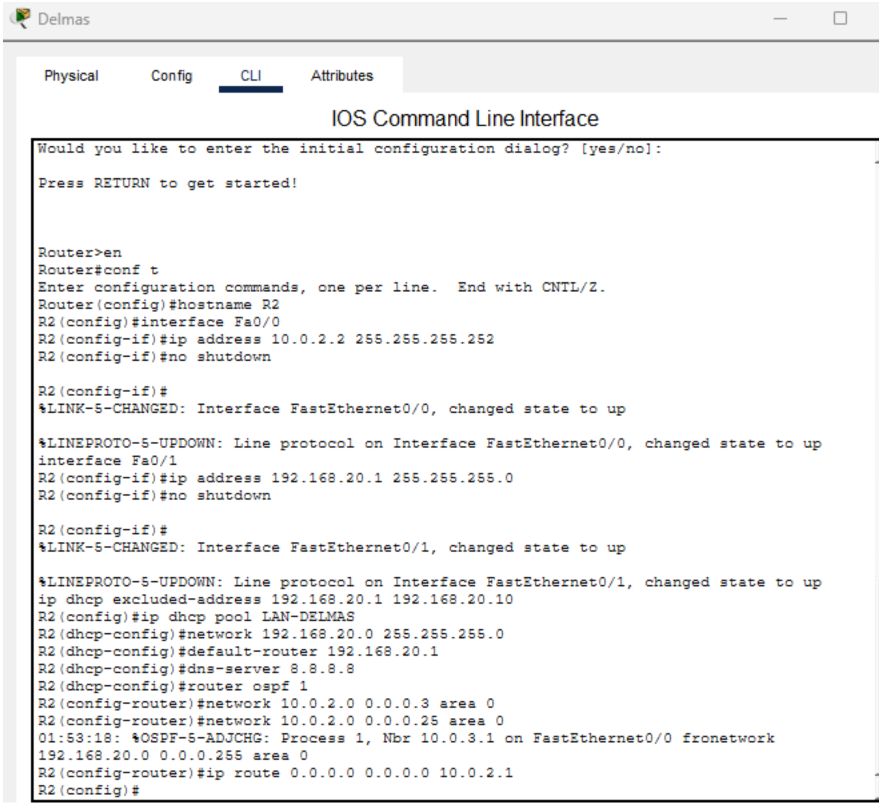
- **Configuration des Routers**

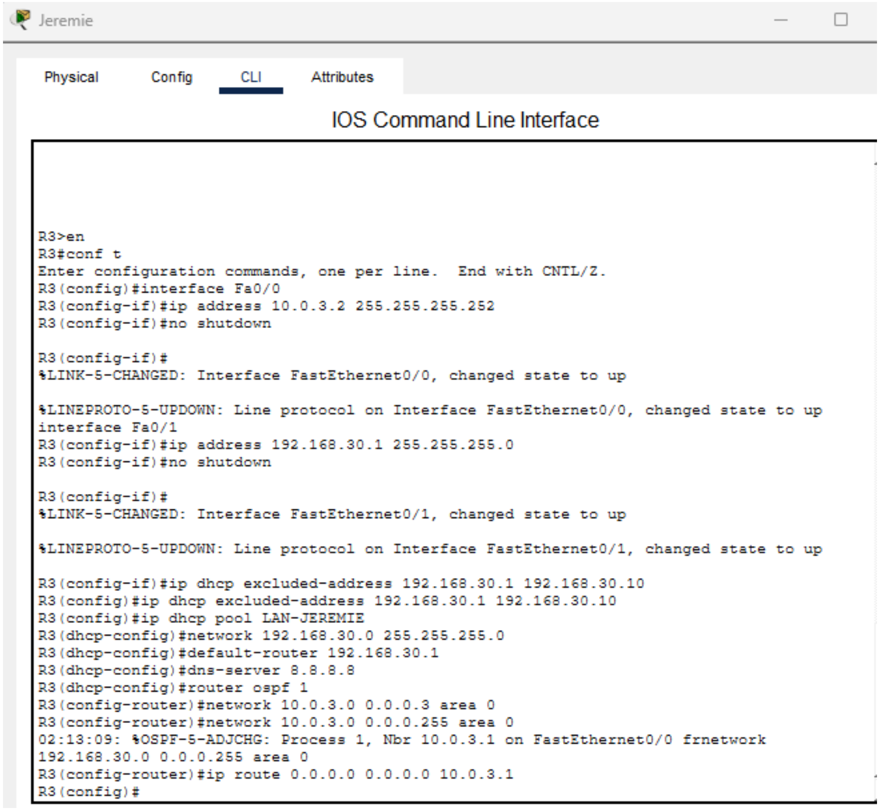
Jacmel

Physical Config CLI Attributes

IOS Command Line Interface

```
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)


         --- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 10.0.1.2 255.255.255.252
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
interface FastEthernet0/1
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip dhcp excluded-address 192.168.10.1 192.168.10.10
R1(config)#ip dhcp pool LAN-JCMEL
R1(dhcp-config)#network 192.168.10.0 255.255.255.0
R1(dhcp-config)#default-router 192.168.10.1
R1(dhcp-config)#dns-server 8.8.8.8
```



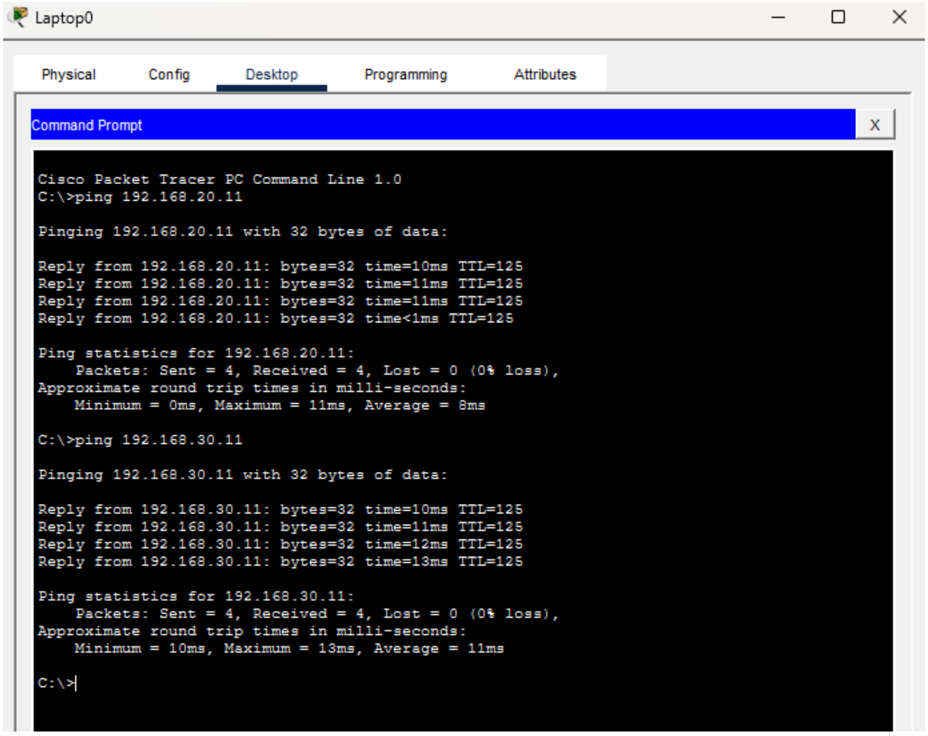


Test

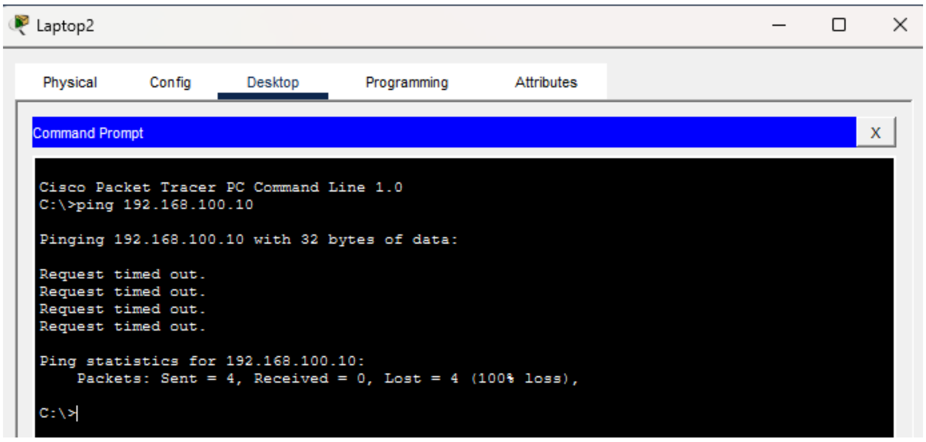
- ✓ **Sur le switch Multilayer**

```
Switch>show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.30.1      1   FULL/BDR        00:00:35    10.0.3.2        FastEthernet0/4
192.168.10.1      1   FULL/BDR        00:00:34    10.0.1.2        FastEthernet0/2
192.168.20.1      1   FULL/BDR        00:00:30    10.0.2.2        FastEthernet0/3
Switch>show ip route ospf
O    192.168.10.0 [110/2] via 10.0.1.2, 00:49:32, FastEthernet0/2
O    192.168.20.0 [110/2] via 10.0.2.2, 00:38:53, FastEthernet0/3
O    192.168.30.0 [110/2] via 10.0.3.2, 00:19:23, FastEthernet0/4
```

- ✓ **Jacmel → ping vers Delmas**
- ✓ **Jacmel → ping vers Jérémie**



- ✓ **Test d'accès à la DMZ pour Jeremie (bloqué par ACL)**




```
R0>ping 203.0.113.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 203.0.113.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R0>
```

```
ciscoasa#show route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
```

```
ciscoasa#show access-list
access-list cached ACL log flows: total 0, denied 0 (deny-flow-max 4096) alert-interval
300
access-list INSIDE-IN; 2 elements; name hash: 0xdae5acda
access-list INSIDE-IN line 1 extended deny ip 192.168.30.0 255.255.255.0 192.168.100.0
255.255.255.0 (hitcnt=0) 0x7a9e844e
access-list INSIDE-IN line 2 extended permit ip any any(hitcnt=0) 0x634d566b
access-list OUTSIDE-IN; 2 elements; name hash: 0x696a7ad3
access-list OUTSIDE-IN line 1 extended permit tcp any host 203.0.113.10 eq www(hitcnt=0)
0xb354080d
access-list OUTSIDE-IN line 2 extended permit tcp any host 203.0.113.10 eq 443(hitcnt=0)
0x2535bd73
```

```
Switch>show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.30.1      1   FULL/BDR        00:00:35    10.0.3.2        FastEthernet0/4
192.168.10.1      1   FULL/BDR        00:00:34    10.0.1.2        FastEthernet0/2
192.168.20.1      1   FULL/BDR        00:00:30    10.0.2.2        FastEthernet0/3
Switch>show ip route ospf
O    192.168.10.0 [110/2] via 10.0.1.2, 00:49:32, FastEthernet0/2
O    192.168.20.0 [110/2] via 10.0.2.2, 00:38:53, FastEthernet0/3
O    192.168.30.0 [110/2] via 10.0.3.2, 00:19:23, FastEthernet0/4

Switch>
```

Partie 2 - Mise en place d'un VPN sécurisé pour le réseau universitaire IUS

Jacmel

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
R1(config)#access-list 120 permit ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes
R1(config-isakmp)#hash sha
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 2
R1(config-isakmp)#lifetime 86400
R1(config-isakmp)#crypto isakmp key cisco123 address 10.0.2.2
R1(config)#crypto isakmp key cisco123 address 10.0.3.2
R1(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R1(config)#
R1(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#match address 110
R1(config-crypto-map)#set peer 10.0.2.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#crypto map VPN-MAP 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#match address 120
R1(config-crypto-map)#
R1(config-crypto-map)#set peer 10.0.3.2
R1(config-crypto-map)#set transform-set VPN-SET
R1(config-crypto-map)#interface FastEthernet0/0
R1(config-if)# crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#
R1(config-if)#
```

Delmas

Physical Config CLI Attributes

IOS Command Line Interface

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
R2(config)#access-list 120 permit ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
R2(config)#crypto isakmp policy 10
R2(config-isakmp)#encryption aes
R2(config-isakmp)#hash sha
R2(config-isakmp)#authentication pre-share
R2(config-isakmp)#group 2
R2(config-isakmp)#lifetime 86400
R2(config-isakmp)#crypto isakmp key cisco123 address 10.0.1.2
R2(config)#crypto isakmp key cisco123 address 10.0.3.2
R2(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R2(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R2(config-crypto-map)#
R2(config-crypto-map)#match address 110
R2(config-crypto-map)#set peer 10.0.1.2
R2(config-crypto-map)#set transform-set VPN-SET
R2(config-crypto-map)#crypto map VPN-MAP 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R2(config-crypto-map)#match address 120
R2(config-crypto-map)#set peer 10.0.3.2
R2(config-crypto-map)#set transform-set VPN-SET
R2(config-crypto-map)#interface FastEthernet0/0
R2(config-if)#crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
```

Jeremie

Physical Config CLI Attributes

IOS Command Line Interface

```
R3>en
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#access-list 110 permit ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
R3(config)#access-list 120 permit ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
R3(config)#crypto isakmp policy 10
R3(config-isakmp)#encryption aes
R3(config-isakmp)#hash sha
R3(config-isakmp)#authentication pre-share
R3(config-isakmp)#group 2
R3(config-isakmp)# lifetime 86400
R3(config-isakmp)#crypto isakmp key cisco123 address 10.0.1.2
R3(config)#crypto isakmp key cisco123 address 10.0.2.2
R3(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
R3(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R3(config-crypto-map)#match address 110
R3(config-crypto-map)#set peer 10.0.1.2
R3(config-crypto-map)#set transform-set VPN-SET
R3(config-crypto-map)#crypto map VPN-MAP 20 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R3(config-crypto-map)#match address 120
R3(config-crypto-map)#set peer 10.0.2.2
R3(config-crypto-map)#set transform-set VPN-SET
R3(config-crypto-map)#interface FastEthernet0/0
R3(config-if)# crypto map VPN-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R3(config-if)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console
```

Test

- **Sur les 3 routers**

✓ Show crypto isakmp sa

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

IPv6 Crypto ISAKMP SA
```

```
R2#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

IPv6 Crypto ISAKMP SA
```

```
R3#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

IPv6 Crypto ISAKMP SA
```

✓ Show crypto ipsec sa

```
R1#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: VPN-MAP, local addr 10.0.1.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   current_peer 10.0.2.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.1.2, remote crypto endpt.:10.0.2.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x0(0)

     inbound esp sas:
```

```
R2#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: VPN-MAP, local addr 10.0.2.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   current_peer 10.0.1.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.2.2, remote crypto endpt.:10.0.1.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x0(0)

     inbound esp sas:
```

```
R3#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: VPN-MAP, local addr 10.0.3.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.30.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   current_peer 10.0.1.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.3.2, remote crypto endpt.:10.0.1.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x0(0)

     inbound esp sas:
```

✓ Show crypto map

```
R1#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
        Peer = 10.0.2.2
        Extended IP access list 110
            access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
        Current peer: 10.0.2.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
        PFS (Y/N): N
        Transform sets={
                VPN-SET,
        }
        Interfaces using crypto map VPN-MAP:
                FastEthernet0/0

Crypto Map VPN-MAP 20 ipsec-isakmp
        Peer = 10.0.3.2
        Extended IP access list 120
            access-list 120 permit ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
        Current peer: 10.0.3.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
        PFS (Y/N): N
        Transform sets={
                VPN-SET,
```

```
R2#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
        Peer = 10.0.1.2
        Extended IP access list 110
            access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
        Current peer: 10.0.1.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
        PFS (Y/N): N
        Transform sets={
                VPN-SET,
        }
        Interfaces using crypto map VPN-MAP:
                FastEthernet0/0

Crypto Map VPN-MAP 20 ipsec-isakmp
        Peer = 10.0.3.2
        Extended IP access list 120
            access-list 120 permit ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
        Current peer: 10.0.3.2
```

```
R3#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
        Peer = 10.0.1.2
        Extended IP access list 110
            access-list 110 permit ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
        Current peer: 10.0.1.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
        PFS (Y/N): N
        Transform sets={
                VPN-SET,
        }
        Interfaces using crypto map VPN-MAP:
                FastEthernet0/0

Crypto Map VPN-MAP 20 ipsec-isakmp
        Peer = 10.0.2.2
        Extended IP access list 120
            access-list 120 permit ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
        Current peer: 10.0.2.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
        PFS (Y/N): N
        Transform sets={
                VPN-SET,
```

Test

✓ Jacmel

```
R1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

IPv6 Crypto ISAKMP SA


R1#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: VPN-MAP, local addr 10.0.1.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   current_peer 10.0.2.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.1.2, remote crypto endpt.:10.0.2.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x0(0)

     inbound esp sas:


R1#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
        Peer = 10.0.2.2
        Extended IP access list 110
            access-list 110 permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
        Current peer: 10.0.2.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
```

✓ Delmas

```
R2#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

IPv6 Crypto ISAKMP SA


R2#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: VPN-MAP, local addr 10.0.2.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   current_peer 10.0.1.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.2.2, remote crypto endpt.:10.0.1.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x0(0)

     inbound esp sas:


R2#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
        Peer = 10.0.1.2
        Extended IP access list 110
            access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
        Current peer: 10.0.1.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
```

- ✓ **Jeremie**

```
R3#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status

IPv6 Crypto ISAKMP SA


R3#show crypto ipsec sa

interface: FastEthernet0/0
    Crypto map tag: VPN-MAP, local addr 10.0.3.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.30.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
   current_peer 10.0.1.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.3.2, remote crypto endpt.:10.0.1.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet0/0
     current outbound spi: 0x0(0)

     inbound esp sas:


R3#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
        Peer = 10.0.1.2
        Extended IP access list 110
            access-list 110 permit ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
        Current peer: 10.0.1.2
```

QUESTIONS

1. Différence entre NAT statique et NAT dynamique

Le NAT statique associe une adresse privée à une adresse publique fixe et permanente, ce qui est idéal pour exposer des serveurs internes (comme un serveur web en DMZ) à Internet. Par exemple, 192.168.100.10 → 203.0.113.10 permet aux utilisateurs externes d'accéder au serveur via l'adresse publique. À l'inverse, le NAT dynamique (ou PAT) permet à plusieurs adresses privées de partager une seule adresse publique (ou un pool) en utilisant des ports uniques. Cela est utile pour permettre aux hôtes internes d'accéder à Internet sans exposer leurs adresses privées. En résumé, le NAT statique est utilisé pour les serveurs, tandis que le NAT dynamique est utilisé pour les réseaux internes.

2. Que fait une ACL sur ASA ?

Une ACL sur ASA filtre le trafic entrant ou sortant sur une interface spécifique (comme inside, outside ou dmz). Elle définit des règles pour autoriser ou bloquer le trafic en fonction des adresses IP, ports ou protocoles. Par exemple, une ACL peut bloquer tout le trafic de Jacmel (192.168.20.0/24) vers la DMZ (192.168.100.0/24) tout en autorisant le reste. Les ACL sur ASA sont étatful, ce qui signifie qu'elles suivent l'état des connexions et autorisent automatiquement le trafic de retour pour les sessions déjà établies.

3. Pourquoi segmenter les réseaux par site ?

La segmentation réseau consiste à diviser un réseau en sous-réseaux distincts pour chaque site. Cela permet de limiter les broadcasts, d'améliorer la sécurité en isolant les sites les uns des autres, et de simplifier la gestion du réseau. Par exemple, en segmentant les sites, vous pouvez appliquer des règles de sécurité spécifiques à chaque segment, comme bloquer le trafic entre Jacmel et la DMZ. Cela facilite également le routage (avec OSPF) et le dépannage, car chaque site peut être géré indépendamment.

4. Pourquoi utiliser une clé pré-partagée ?

Une clé pré-partagée est une méthode simple et rapide pour authentifier les extrémités d'un tunnel VPN IPsec. Elle est idéale pour les environnements de laboratoire ou les petits réseaux, car elle ne nécessite pas d'infrastructure complexe comme une PKI (Public Key Infrastructure). Cependant, elle est moins sécurisée qu'un certificat, car si la clé est compromise, un attaquant pourrait établir une connexion VPN non autorisée. Dans votre TD, la clé cisco123 est utilisée pour authentifier les routeurs entre eux.

5. Que fait une ACL dans un VPN ?

Dans un VPN IPsec, une ACL définit le trafic "intéressant" qui doit être chiffré et envoyé à travers le tunnel. Par exemple, une ACL peut spécifier que tout le trafic entre Port-au-Prince (192.168.10.0/24) et Jacmel (192.168.20.0/24) doit être chiffré. Sans cette ACL, le routeur ne saurait pas quel trafic protéger, et le VPN ne fonctionnerait pas correctement. L'ACL est référencée dans la crypto map, qui lie le trafic intéressant à une politique de sécurité IPsec.

6. Pourquoi chiffrer les communications inter-sites ?

Le chiffrement des communications inter-sites est essentiel pour protéger les données sensibles (comme les notes des étudiants ou les informations administratives) lorsqu'elles transitent sur un réseau non sécurisé comme Internet. IPsec garantit la confidentialité (chiffrement AES), l'intégrité (HMAC-SHA) et l'authenticité (ISAKMP) des données. Cela empêche les attaquants d'intercepter, modifier ou usurper les communications. De plus, le chiffrement permet de respecter les réglementations comme le RGPD ou le FERPA.

7. Différence entre Phase 1 et Phase 2 dans IPsec.

La Phase 1 (ISAKMP) établit un canal sécurisé entre les deux extrémités du tunnel VPN pour négocier les paramètres de sécurité (comme le chiffrement, le hachage et l'authentification). Elle aboutit à la création d'une SA IKE (Security Association), qui est bidirectionnelle. La Phase 2 (IPsec) utilise ce canal pour négocier les tunnels de données qui chiffreront effectivement le trafic utilisateur. Elle crée des SA IPsec, qui sont unidirectionnelles et ont une durée de vie plus courte que celles de la Phase 1. La Phase 2 est déclenchée par le trafic intéressant défini par une ACL. Les deux phases sont complémentaires : la Phase 1 sécurise la négociation, tandis que la Phase 2 protège les données.

Conclusion

Ce travail m' a permis de mettre en place une infrastructure réseau sécurisée pour l'université IUS, répartie sur trois sites : Jacmel, Delmas et Jérémie.